

Fragmenting Space and Time: Bridging Kinetic Sculptures and Pre-Cinematic Concepts for Virtual Reality

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ABSTRACT

ExperiMotion 3, part of the *VRinMotion* project, collaborated with artist Anna Vasof to explore the fusion of historical media and modern virtual reality technologies. The project included two key concepts: capturing photo sequences of a bird's flight using a VR camera inspired by Marey's chronophotographic gun, and a moving bird-cage sculpture where visitors, wearing a VR headset, experienced an illusion of linear time via space fragmentation and stop-motion animation. The research addressed questions about extending stop-motion techniques to VR, merging physical objects with immersive experiences, creating time illusions, and exploring novel interaction methods like eye-blinking, while also considering the role of VR users as both participants and observed protagonists.

Keywords: Virtual Reality, Stop-Motion Animation, Art-Based Research, Virtual Live Performance

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Arts and humanities—Media arts

1 INTRODUCTION

VRinMotion, an art-based research project, combines various stop-motion techniques in VR setups. The project integrates artistic virtual experiences with philosophical, theoretical, and analytical concepts. Through four experimental workshops, the *VRinMotion* team collaborates with animation artists, collaborators, and students to expand artistic concepts of stop motion and motion capture in virtual and hybrid environments. Its third iteration, *ExperiMotion 3*, took place in collaboration with guest artist Anna Vasof, whose artistic practice develops new mechanisms of cinematography and animation to produce critical films, actions and installations (see [27]). Her collaboration with the *VRinMotion* team involved exploring the relationship between real objects and their virtual counterparts, as well as the manipulation and representation of the space-time continuum within virtual reality using stop-motion animation and eye-tracking.

Two different project ideas were conceived and implemented to varying degrees of completeness. Both ideas dealt loosely with parallels between historical media and modern technologies. In

the first implementation, a protagonist was encouraged to capture photo sequences of a bird's flight in a virtual environment using a virtual camera, which, however, had a historical counterpart, as it was based on Étienne-Jules Marey's chronophotographic gun. When the virtual camera was triggered, a digital photo and a rough 3D representation of the protagonist from outside the VR were also taken. These captured individual images could then be reassembled into an animated sequence. A moving birdcage sculpture was created for the second part of *ExperiMotion 3*. Using a 360-degree camera in the cage, visitors can experience the view from within the cage by wearing a VR headset to immerse themselves in a visual and sonic illusion of linear time by means of space fragmentation and stop-motion animation, while closing an eye interrupts the illusion.

In the project's art-based research process, the following questions were developed:

1. How can pre-cinematographic concepts of stop motion be extended to virtual environments?
2. How can a tangible presence of (kinetic) objects be merged with the immersive experience of a VR headset?
3. How can a visual and sonic illusion of time be created using space fragmentation and stop-motion animation?
4. How can unconventional interfaces for virtual reality, such as eye-blinking, create novel interaction modalities?
5. What does it mean for VR users to see themselves within the VR experience while being observed as the protagonists of the VR experience by viewers in the real world?

2 STOP MOTION

Stop motion is a key analogue animation technique since its introduction by Georges Méliès in 1896. It has been used in various forms, including cutout, silhouette, pixilation, and puppet animation, playing a key role in both children's television and experimental art [8] [7]. In recent years, it has been merged with digital techniques [18], [19], [30]. Independent animators have also used stop motion to explore movement and temporality [8, p. 32]. In general, stop motion vitalizes real objects in a fragmented way and is therefore an ideal concept for creating and expanding a new artistic understanding of perception. Viewed from this perspective, stop motion functions not as a traditional animation technique but as a *pre- and post-cinematographic* concept and becomes a tool to fragment and recompose the world. This deconstruction and fragmentation of an object in discrete units, and a reconstruction into an unprecedented entity, opens up new realities.

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2.1 Artistic Input

Media artist and architect Anna Vasof investigates in her project *Non Stop Stop Motion* the essence of the magic trick of cinematography by transforming everyday situations, objects, spaces and actions as cinematographic mechanisms (see [27]). Her films *Domino* [22] and *Self-Portrait* [25] focus not only on multiple observations of everyday life, but also on inventions within it. *Non Stop Stop Motion* is a concept that aims to become a new method that could reveal unexpected narratives and understandings of how we perceive the things we see:

“Sometimes in order to create a Non Stop Stop Motion film, I set up a sequence of objects or images, which I name hyperframes. Hyperframes are the single units, objects or images, which form the sequence and cause the illusion of movement in Non Stop Stop Motion films. I use the term hyperframes in order to differentiate them from the frames that the video camera records (for example a standard video camera records 25 frames per second). Usually more than one video frame is required to record a hyperframe. Video frames reveal what happens in between the hyperframes and connect the viewer with the actual recorded space and action [28].”

Machine [23], by Vasof, is an example of a performative action that creates a stop-motion fiction that comes into conflict with the resulting performance. In *Trilogy of Leaving* [26] and *Travel to the Window* [24] the process of making the film becomes part of the film, in order to not only reveal the technique but also to combine the animated story with the causing action. This makes *Non Stop Stop Motion* the ideal medium for stories containing irony and references or metaphors for personal or social conflicts.

2.2 Pre- and Post-Cinema

Various researchers and artists have experimented with intermediate expansion of animation in combination with pre- and post-cinematic concepts. Julien Maire created a so-called *stereolithographic* projector which projects the shadows of 3D printed sculptures attached to a film strip. As a conceptual irony about 3D cinema, *Man at Work* [17] contains an endless animation of a man digging into the ground. The small, fragmented movements of *Man at Work* not only animate the sculptures, but extend the cinematic understanding into time and space. Akinori Goto created a 3D-printed zoetrope in 2016, which uses light projection to bring moving figures to life (see [10]). This technology allows him to create an infinite number of frames/fragments, generating the absolute simultaneous and sequential experience of movement. A similar concept was established by John Edmark for his strobe animated sculptures *Blooms* [5] and *Blooms 2* [6]. Unlike a 3D zoetrope, which creates the illusion of small shape changes through a series of objects, *Blooms* are animated as single, independent sculptures. There have also been some efforts to replicate the functionality of a classical zoetrope within virtual reality environments [12], [16], [9]. In [14] the expansion of animation through the combination of light, shadow, and historical techniques was explored in an artistic research project. Two of the experiments conducted in the project (*Artifact 4* and *Artifact 5*) integrate proto-animation devices like the zoetrope and phenakistoscope into an immersive environment. Using VR and 3D printing, the works engage viewers with animated shadow projections, inspired by expanded cinema traditions. The generative VR experience *Running Nude* [21] by Vladimir Todorovic is inspired by Étienne-Jules Marey's chronophotographs and Marcel Duchamp's *Nude Descending a Staircase*. The project reimagines human locomotion in a VR space enhanced by AI storytelling. Users are faced with figures whose limbs and bodies morph into soft, elastic, and easily deformable forms.



Figure 1: Guest artist Anna Vasof painting the 3D-printed model of a super 8 camera equipped with a mechanism to be clipped to a Quest 2 controller.

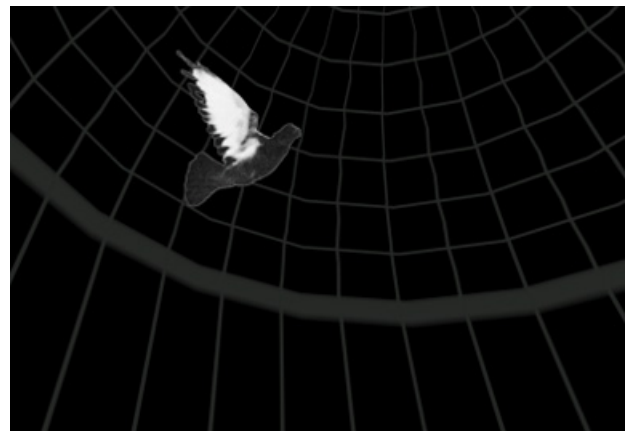


Figure 2: Isolated image of a single bird in flight taken from [15] displayed in the VR environment build by the VRinMotion team under the direction of Anna Vasof.

In line with the pre- and post-cinematographic concepts mentioned above, the experiments with 3D stop motion in *ExperiMotion 3* were based on two prominent pre-cinematographic developments: the motion studies of Étienne-Jules Marey and Thomas Alva Edison's kinetoscope. Marey's photographs of individual phases of movement, realized with the aid of serial photography, can be seen as a direct precursor to stop-motion animation [20]. After all, Marey only had to wait for the development of celluloid film to reconstruct the original movement from its frozen snapshots [13]. Edison's kinetoscope represents a historical development on the road to cinematography that would soon prove to be a dead end. The kinetoscope could create a perfect illusion of motion, but only for one viewer at a time. It may have offered a greater immersion in the motion displayed than the projection of moving images in cinemas, which was to follow the kinetoscope only a short time later. However, the precondition for this immersion, which was the isolation of the viewer in front of the apparatus, was also the main obstacle to the kinetoscope's mass appeal. The situation with immersion in virtual worlds is currently very similar: the isolation of the viewer associated with wearing VR glasses (in addition to their weight) seems to be a major factor in their failure to really take off as a mass consumer item.



Figure 3: Protagonist of the performance captured with the system outlined in 3.1.

3 METHODS AND OUTCOME

In contrast to the methodical and rational approach of science and technology, Anna Vasof's artistic endeavours always have a playful dimension. The approach is not linear, but more like a board game in which different elements are combined until something interesting and unpredictable emerges. In the case of *ExperiMotion 3*, the elements were, on the one hand, the reference to pre-cinematographic devices and, on the other, the reflection of the experience associated with VR. Vasof's approach coincides with that of *VRinMotion*, which is based on Borgdorff's framework of artistic research: Research on the Arts, Research for the Arts, and Research in the Arts (see [2]). All developments were documented through logs, videos, voice notes, and screen captures, offering valuable insights into the decision-making process and key milestones. This documentation helps understand the challenges of combining artistic, theoretical and technological methods in VR research and provides transparency into the artistic and research processes.

3.1 Animation Setup for *Shooting Marey*

A central concept developed during the project is inspired by one of Étienne-Jules Marey's motion studies ([Bird in Flight] [15]) (see figure 2). Participants use a VR controller outfitted with a 3D-printed model of a Super-8 camera to photograph virtual birds (see figure 1). The camera includes a mechanism that allows it to be clipped into the circular section of a Quest 2 controller. Pressing the trigger button on the controller simultaneously captures images of the participant from the outside using both an Azure Kinect and a DSLR camera.

Vasov is generally interested in the perspective of external observers watching individuals engaged in VR experiences. The VR environment is a minimalistic black space featuring a hemispherical grid reminiscent of a birdcage, with additional grid lines on the floor. Within this space, users operate a virtual camera (a 3D model visible within the VR environment) searching for and capturing birds through its viewfinder. The birds are represented as individual frames from Marey's motion studies, appearing sequentially in the VR environment. Each photograph taken in VR triggers an external image capture, documenting the participant as they rotate and interact with the virtual scene. The resulting sequences of images offer two different perspectives: the inside perspective, consisting of the bird images captured in VR, and the outside perspective, composed of images of the user's physical movements. The inside perspective offers a view of the user's task and its aesthetic focus, while the outside perspective reveals the abstract and often disjointed view of someone engaging with an invisible virtual world. Played back as loops, these sequences contrast the immersive experience of VR



Figure 4: View from outside the *The Cage of Time*.

with the physical reality of its use, offering a commentary on the layered experiences of virtual interaction and spectatorship. It is also worth noting that in a performance realized according to the artist's vision using the created system, the protagonist wears a historical costume as a quasi-reference to the era of Étienne-Jules Marey (see figure 3). This choice is intended to bridge the two layers: the virtual reality experience and the external perspective on the individual.

3.2 Implementation into VR for *Shooting Marey*

For the technical implementation, the *Unity Interaction Toolkit* was used in the project to enable basic VR interactions, such as grabbing and aligning virtual objects. A 3D model of a Super 8 camera was integrated into VR, equipped with a virtual camera on its lens. When the camera shutter is activated, a raycast is emitted to detect the photographic subjects. The virtual camera is assigned a render texture, displayed and magnified in the viewfinder of the Super 8 model. The photographic subjects are chronophotographs of birds, integrated as a sprite sheet (see figure 2). These sprites are sequentially placed in the environment, instantiated randomly on a hemisphere at a fixed distance from the viewer. Each successful capture saves the virtual photo locally, removes the current subject, and instantiates a new one. Additionally, an OSC signal is sent to Dragonframe to trigger a DSLR camera, capturing the participant's interaction. The 3D model of the Super 8 camera was modified in *Maya* to include a *Meta Quest Pro* controller mount, 3D printed, painted, and decorated with stickers (see figure 1).

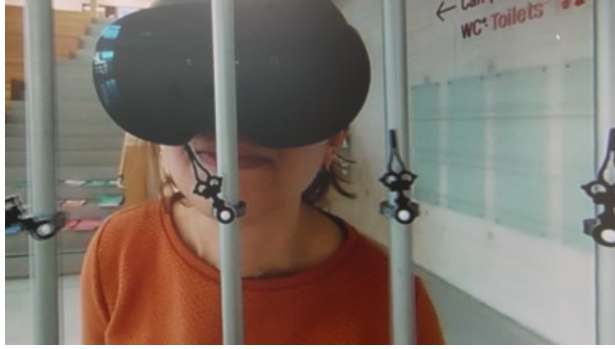


Figure 5: View from inside the *The Cage of Time*.

3.3 Animation Setup for *The Cage of Time*

The experience of making *Shooting Marey* brought another element into play: the hemispherical grid of the virtual space (see figure 2). This grid inspired Vasof to recreate it as a moving object in reality. This not only shifted the fundamental orientation of the first experiment with space and movement towards a reflection of time. What would later be called *The Cage of Time* also provided an elegant solution to the problem of merging the internal and external perspectives of the virtual experience.

The Cage of Time is a rotating object with a circular form, featuring 16 clock pointers mounted on the bars of the cage in succession (see figure 5), reminiscent of a pre-cinematic device. The rotation of the clock pointers *operates* on similar principles as a pre-cinematic animation, but here, the separation and succession of the frames occur digitally, resulting in a side effect: the fragmentation of the surrounding space. This affects the perception of continuity in VR media, multiplying the sense of spatial dislocation. This dislocation, combined with the impression of seemingly continuous time, calls our spatial existence into question.

The object itself is a kinetic sculpture in the form of a birdcage. It continuously rotates the bottom part and produces sound. The bottom part is made of a drum cymbal transformed into a clock panel with Latin numbers. When it rotates, it hits the cage bars that move slightly and produce a metallic sound. Static clock pointers are mounted on each of the cage bars. In the middle of the cage, a 360° camera is mounted and constantly records the space surrounding the cage and sends the footage to a computer; with the help of software, it fragments the space and creates the animated illusion of the rotation of the clock pointers. The audience can experience this illusion of continuous time by wearing a VR headset. In other words, when exhibition visitors wear the headset, they are teleported into the inside of the cage, experiencing the space fragmented and simultaneously seeing the rotation of the clock pointers. The illusion of passing time in this project is literal, as we use stop motion to achieve it, and as a result, we fragment the continuous perception of space.

By closing one eye, the audience can interrupt the illusion of time and restore the feeling of continuous space and duration. A microphone from inside the cage also transfers the sound. With the help of software, the sound is also slightly manipulated, amplifying the claustrophobic and dizzy feeling of *The Cage of Time*.

3.4 Implementation into VR for *The Cage of Time*

The technical setup (see figure 6) consists of a 360° camera that live-streams its video output via USB to a computer running VVVV [11] for viewing in a VR environment with 3 degrees-of-freedom. A *Meta Quest Pro* VR headset is used for display, as it also provides eyetracking data. The VR headset is connected wirelessly with

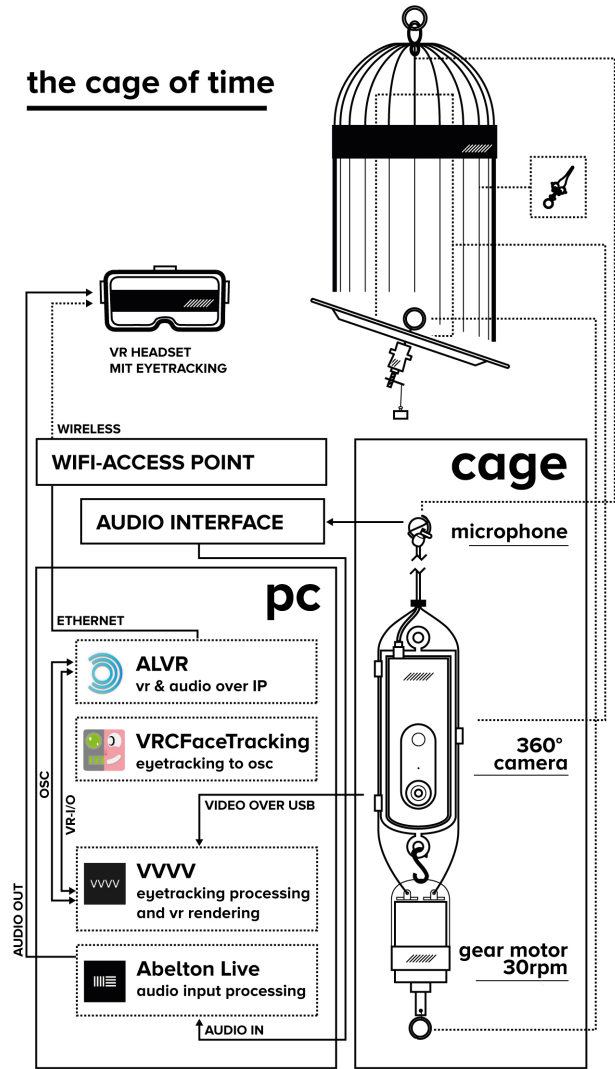


Figure 6: Schematic representation of the hardware and software used in *The Cage of Time*.

the PC, using *ALVR* [3] as a connection and streaming handler. A customized version of *VRCFaceTracking* [4] hooks into *ALVR* and converts the eye tracking data to *OSC*. A microphone inside the cage provides an audio input, which is processed in *Ableton Live* [1] with a chain of filters, which themselves are again controlled from the eye-tracking data. The processed audio output is sent back to the VR headset using *ALVR*.

4 DISCUSSION AND CONCLUSION

Both, *Shooting Marey* and *The Cage of Time* referred to pre-cinematic concepts, to Étienne-Jules Marey's chronophotographic gun and to Thomas Edison's kinetoscope. However, embedding them in a VR environment produced very different results: While *Shooting Marey* only loosely referenced one of the original stop-motion scenes and shifted the dimension of motion analysis to the observation of the person experiencing VR, *The Cage of Time* not only created a kinetoscope-related illusion of motion, but also established a reference to the fair as a place where illusions of motion were offered. The cage's ability to capture the audience's attention as

an enigmatic kinetic object was demonstrated at the first exhibition of *The Cage of Time* at the 19th Best Austrian Animation Festival in the Semperdepot, Vienna. The work was then invited to the prestigious *FILE Festival* in Sao Paolo, where it was shown from July 2nd to the 25th of August 2024. Throughout the two exhibitions, the discrepancy between viewing the cage from the outside and the virtual experience of being caught up in the dizzying motion inside the cage not only regularly amazed audiences, but also led to numerous reflections on the immersive nature of the virtual experience.

For Anna Vasof, the ability to see oneself from inside the cage and the option to stop the motion illusion by closing one eye were particularly important anchor points for breaking out of the immersive experience in VR:

“By blurring the boundaries between physical and digital worlds, *The Cage of Time* not only offers a captivating experience but also prompts reflection on the nature of perception, reality, and the passage of time itself. Through this interactive installation, viewers are encouraged to contemplate the sound and the fluidity of existence and the interconnectedness of the tangible and the intangible [29, p.1].”

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